Abnormal Object Detection by Canonical Scene-based Contextual Model

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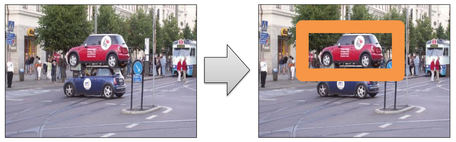
INTRODUCTION

Problem Statement

■ Abnormal Object Detection (AOD)



Input Output

➢ If an object rarely co-occurs with other existing objects or being floated/sunken/big/small compared with normal objects, it is classified abnormal object.

■ Motivation

➢ Increasing number of abnormal images which are generated by image manipulation tools or artistic photographers

➢ Applicable to visual surveillance pipeline

Previous Works & Limitations

Tree-based model
Tree-relation among obj. or obj. inst.

LESS affluent object relation

M. J. Choi, A. Torralba, and A. S. Willsky, *Context Models and Out-of-context Objects*, Pattern Recognition Letters, 2012.

Our Approach

Latent scene/camera-variable model
Richer relation among obj. or obj. inst.

MORE affluent object relation

Our Contribution

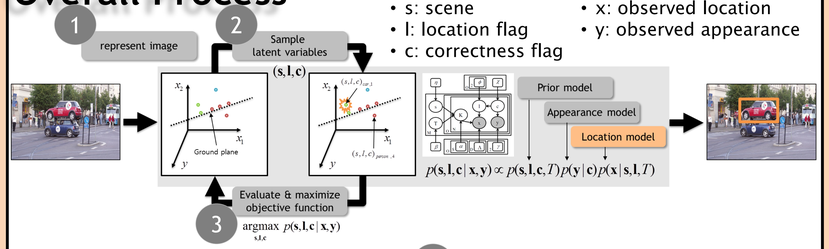
Solve new emerging vision problem

Novel Latent model

New abnormal dataset annotated in object-level

PROPOSED METHOD

Overall Process

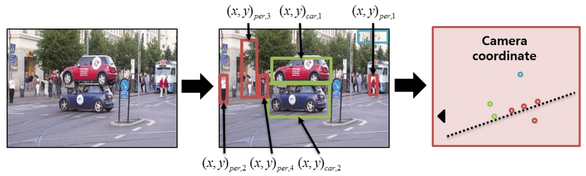


- s: scene
- l: location flag
- c: correctness flag
- x: observed location
- y: observed appearance

Image Representation 1

Object-level image rep.

"Undo" projectivity



➢ Represent image by a set of bounding boxes that are extracted by object detectors

➢ Each image consists of $\sum_{o=1}^{O} N_o$ bounding boxes (=100, in this paper)

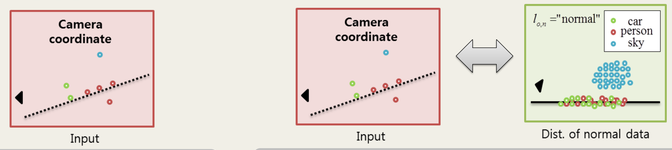
➢ Transform "image coordinate" to "camera coordinate" by simple triangulation

➢ Represent position and scale information altogether

Hoiem, D., Efros, A., Hebert, M.: Putting objects in perspective. *Proc. of IEEE conf. on CVPR*, 2006.

Main Idea

Identify abnormal ones!



Which object is abnormal?

➢ Which object is less co-occur, floated/sunken, or big/small?

Define dist. of normal data & Compare?

➢ Compare the input with the distribution of normal objects

➢ Check likelihood of input given the dist.

➢ How to represent the distribution of normal scene?

⇨ Construct the Canonical Scene (CS) model

➢ How to compare the input scene with the normal scene?

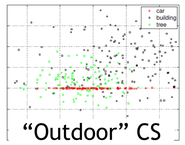
⇨ Matching transformation T for CS

⇨ Similarity measure to compare the input scene and transformed CS

Modeling 3

■ Define "Canonical Scene" $\mathcal{L}_s = \{L_{o,n,d}\}$

Canonical Scene (CS)



"Outdoor" CS

➢ Natural distributions of normal objects

➢ Less co-occurring objects does **not exist**

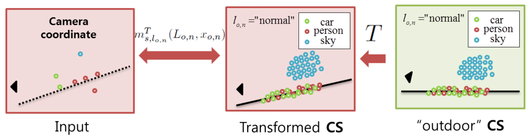
➢ "Objects" are **on the ground plane**

➢ Follows leaned truncated Gaussian distribution

■ Define matching transform. & similarity measure

Matching transform

➢ T: 2D isometric transformation



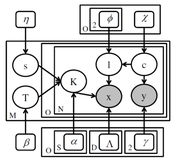
Similarity measure $m^T_{s,l_{o,n}}(L_{o,n},x_{o,n}) \equiv p(K_{o,n},x_{o,n}\,|\,s,l_{o,n},T)$

➢ Defined as multivariate truncated normal distributions

■ Return to the Goal

$$\hat{s},\hat{\mathbf{l}},\hat{\mathbf{c}} = \underset{s,\mathbf{l},\mathbf{c}}{\arg\max} \int \underbrace{p(s,\mathbf{l},\mathbf{c},T|\mathbf{x},\mathbf{y})}_{\text{Model}}\mathrm{d}T$$

Decompose



$p(s,\mathbf{l},\mathbf{c},T)$
Prior model
➢ Prior on latent variables

$\int p(\mathbf{K},\mathbf{x}\,|\,s,\mathbf{l},T)\mathrm{d}\mathbf{K}$
Location(Contextual) Model
➢ Defined by previous similarity measure

$p(\mathbf{y}\,|\,\mathbf{c})$
Appearance Model
➢ Defined as conventional model

Inference 2

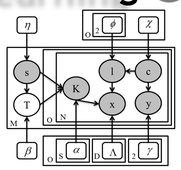
■ Advantage of Pop-MCMC

➢ Multiple Markov chains with genetic operations

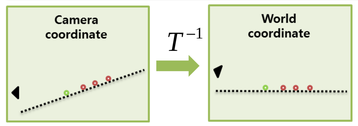
Efficiently escape from local optimum

Kim, W., Park, J., Lee, K.: Stereo matching using population-based MCMC. IJCV 83, 2009.

Learning 3



Estimate T → make complete data



➢ Assumes all "objects" in normal images are on the ground plane

➢ ML-variant learning

EXPERIMENTS

■ Project site opened

➢ http://cv.snu.ac.kr/research/~CSAOD

➢ Dataset and source code are available

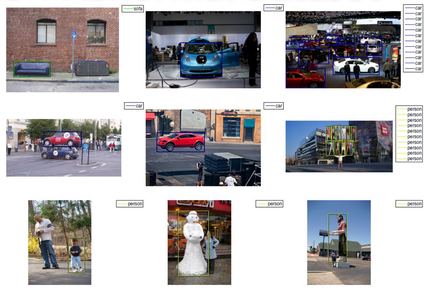
■ Experiment settings

➢ Felzenszwalb's part-based object detectors

➢ 10 object categories

➢ Two scene categories (indoor/outdoor)

New Abnormal Dataset

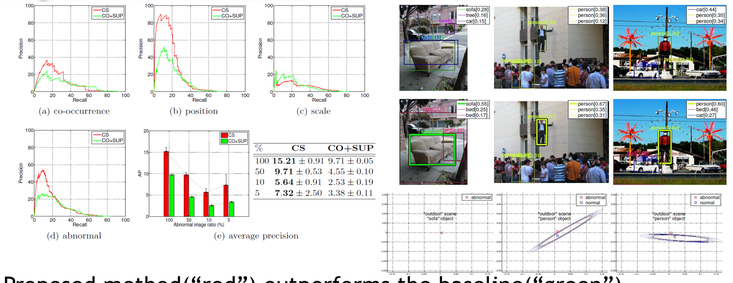


# Abnormal Images	149
Co-occurrence violation	38
Position Violation	53
Scale violation	44
Mixed violation	14

➢ Only abnormal objects are annotated

➢ Scene types are also annotated

Comparison Results

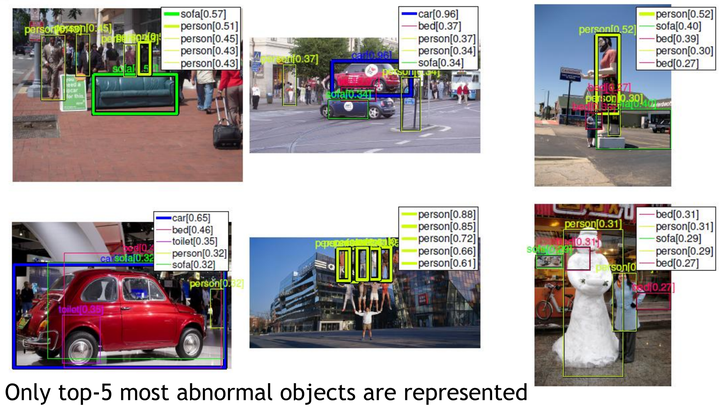


%	CS	CO+SUP
100	$\mathbf{15.21} \pm 0.91$	9.71 ± 0.05
50	$\mathbf{9.71} \pm 0.53$	4.55 ± 0.10
10	$\mathbf{5.64} \pm 0.91$	2.53 ± 0.19
5	$\mathbf{7.32} \pm 2.50$	3.38 ± 0.11

➢ Proposed method("red") outperforms the baseline("green")

➢ Because of affluent object relation, floating person is detected as most abnormal objects

Additional Results



➢ Only top-5 most abnormal objects are represented

CONCLUSION

We proposed a **novel** generative model for **abnormal object detection** that gives the **state-of-the-art** performance on **newly** constructed **abnormal image dataset**.

Full parameter learning and annotation error handling will further improve the results.